

Stage A — Statistical Model

Naive Model:

$$X_1, \dots, X_J \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{1, \dots, I\} \implies Y_i \approx \text{Pois}\left(\frac{J}{I}\right)$$

New Model:

$$Y_k \sim \text{Pois}(\lambda_k), \quad \lambda_k = a_k f(\text{gc}_k) \varepsilon_k$$

Variance Decomposition:

$$\text{Var}(Y_k) = \underbrace{\mathbb{E}[\text{Var}(Y_k \mid \lambda_k)]}_{\mathbb{E}[\lambda_k] \text{ (within-bin Poisson noise)}} + \underbrace{\text{Var}(\mathbb{E}[Y_k \mid \lambda_k])}_{\text{Var}(\lambda_k) \text{ (between-bin heterogeneity)}}.$$

Noise Decomposition:

$$Y_k^T = a_k \cdot f^T(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^T + \eta_k^T, \quad Y_k^N = f^N(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^N + \eta_k^N$$

γ_k = shared noise; δ_k = sample-specific noise; η_k = Poisson additive noise.

Stage B — GC Effect Modeling

GC Bias Function (estimated by cubic B-spline):

$$\mathbb{E}[Y_k \mid \text{gc}_k] = f(\text{gc}_k)$$
$$\hat{f}(\text{gc}) = \hat{\beta}^\top \mathbf{b}(\text{gc}), \quad \mathbf{b}(\text{gc}) = \text{bs}(\text{gc}, \text{knots} = Q_{0.25}, Q_{0.50}, Q_{0.75}, \text{degree} = 3)$$

Stage C — Estimators

Plug-in estimator:

$$\hat{a}_k = \frac{Y_k}{\hat{f}(\text{gc}_k)} \cdot \hat{M}, \quad \hat{M} = \frac{1}{\text{median}_k \left\{ Y_k / \hat{f}(\text{gc}_k) \right\}}$$

Stabilized estimator ($c = 1$):

$$\hat{a}_k = \frac{Y_k + c}{\hat{f}(\text{gc}_k) + c}$$

Two-sample ratio (headline method):

$$\hat{a}_k^{\text{ratio}} = \frac{\frac{Y_k^T + c}{\hat{f}^T(\text{gc}_k)}}{\frac{Y_k^N + c}{\hat{f}^N(\text{gc}_k)}} \approx \frac{a_k \cdot \delta^T}{\delta^N}$$

where shared technical noise γ_k cancels in the ratio.